🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.EE.3

Simplify Algebraic Expressions

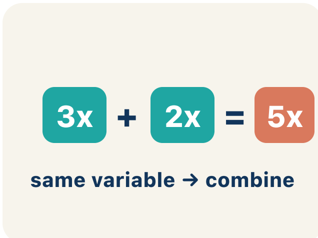
Lesson 6-7 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

Key Vocabulary

Level 2 Standard

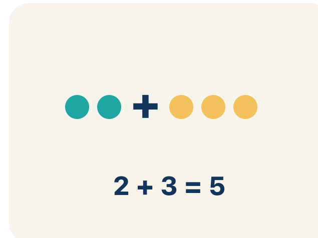
Picture first, then the word, then a plain-language meaning. Say each word out loud.



$5x$ and $3x$ are like terms (both x); $5x$ and $5y$ are NOT (different variables); $2x$ and $2x^2$ are NOT (different powers)

Like Terms

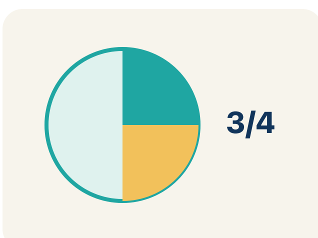
Write the definition:



$8m + 3m = 11m$ — add the coefficients ($8 + 3 = 11$) and keep the variable (m)

Combine

Write the definition:



$2a + 5a + 3$ simplifies to $7a + 3$ — the two a-terms combine, the constant stays

Simplify

Write the definition:

In $9x$, the coefficient is 9 — it tells you 'nine groups of x'; if $x = 2$, then $9x = 18$

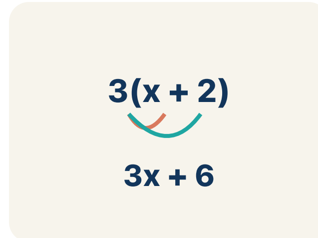
Coefficient

Write the definition:

In 3x + 5 − 2x, the terms are 3x, 5, and 2x.

Term

Write the definition:



3(x + 4) = 3·x + 3·4 = 3x + 12.

Distributive property

Write the definition:

Guided Notes
Level 2 Standard

🎯 **WHAT WE'RE LEARNING TODAY**

I can simplify algebraic expressions by combining like terms.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Like Terms | Combine | Simplify | Coefficient | Term | Distributive property

👆 *Tap any word to see what it means and a picture.*

1. Terms with the same variable part, like 3y and 8y, are ____________.

2. To add or subtract like terms into a single term is to ____________ them.

3. To write an expression in its shortest equal form is to ____________ it.

4. The number multiplied by a variable is the ____________.

5. A number or variable separated by + or – signs is a ____________.

6. Multiplying each part of a sum by a number uses the ____________ property.

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: Simplify 7x + 3x.

A. 10x

B. $10x^2$

C. 21x

D. 73x

Step 1 7x + 3x = 10x.

Step 2 Add the coefficients: 7 + 3 = 10.

✅ **Answer:** A. 10x

🧩 **Try — put the steps in order**

Drag the cards (or use the ▲▼ buttons) to put the solution steps in the right order, then press **Check**.

Add the coefficients: 7 + 3 = 10.

7x + 3x = 10x.

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Simplify 9n + 4 + 2n.

A. 11n + 4

B. 15n

C. $11n^2 + 4$

D. 92n + 4

Step 1 ____________________

Step 2 ____________________

Answer: ____________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Which pair are like terms?

A. $6x$ and $2x$

B. $6x$ and $6y$

C. $6x$ and 6

D. $6x$ and $2x^2$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Clean up this channel: $7y - 3y$.

A. $4y$

B. $10y$

C. 4

D. $21y$

Show your work:

2. Simplify the mix: $6m + 4 + 3m$.

A. $9m + 4$

B. $13m$

C. $9m + 4m$

D. $10m$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Three students simplified $4x + 3 + 2x + 5 + x$ differently. Student A got $7x + 8$. Student B got $7x^2 + 8$. Student C got $6x + 9$. Who is correct? Explain the mistakes the other two students made.

Sentence starter: Student ____ is correct because $4x + 2x + x =$ ____x and $3 + 5 =$ ____. Student ____'s error was ____. Student ____'s error was ____.

Show your work:

Reflect — Exit Ticket

Simplify $6x + 3 + 2x + 5$.

A. $8x + 8$

B. $8x^2 + 8$

C. $62x + 35$

D. $16x$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Like Terms
2. **Notes 2:** Combine
3. **Notes 3:** Simplify
4. **Notes 4:** Coefficient
5. **Notes 5:** Term
6. **Notes 6:** Distributive property
7. **We Try (notes):** A. 11n + 4 *— 9n + 2n = 11n. The constant 4 has no like term, so it stays. Result: 11n + 4.*
8. **You Try (notes):** A. 6x and 2x *— 6x and 2x are like terms because they both have the same variable (x) raised to the same power (1).*
9. **Try It 1:** A. 4y *— Subtract the coefficients of the like terms: 7 − 3 = 4. Keep the variable y, giving 4y.*
10. **Try It 2:** A. 9m + 4 *— 6m + 3m = 9m. The constant 4 has no like term, so it stays. Result: 9m + 4.*
11. **Exit Ticket:** A. 8x + 8 *— 6x + 2x = 8x and 3 + 5 = 8, so the simplified expression is 8x + 8.*